

Janus Resilience Lab

An independent reimplementation, validation, and analysis platform
for the Project Janus collapse–recovery model

Janus Observatory experiments track

August 21, 2026 · dataset jrl-dataset-1.0.0

Abstract

This report documents an experiments-track platform built around the published collapse–recovery model of JANUS-PAPER-05 ([arXiv:2604.13774v1](#), CC BY 4.0). The model is reimplemented independently from its equations in three mutually validating representations - exact renewal recursions, a year-by-year simulator, and an event-driven sampler - and validated against every aggregate the source reports in prose. On that foundation the experiment adds censoring-aware survival analysis, variance-based global sensitivity analysis, a 65,536-run continuous-parameter dataset, a surrogate benchmark under progressively harder holdout protocols, and the paper’s technosignature observation-probability equation applied to reimplemented duty cycles. The ten Project Janus scenarios are self-consistent possibilities, not forecasts: nothing here ranks them, assigns them probabilities, or implies they are equally likely.

Contents

1	The published model	2
2	Replication of published outcomes	2
3	Censoring-aware survival analysis	2
4	Global sensitivity analysis	3
5	Level-2 continuous-parameter dataset	3
6	Surrogate benchmark under grouped holdouts	4
7	Technosignature persistence and observation probabilities	4
8	Adjacent NASA context data	5
9	Limitations and threats to validity	6
10	Provenance and rights	7
A	Reproducing every number and figure	8

1 The published model

JANUS-PAPER-05 models a technological civilization through two state variables - technological capacity $T(t)$, initialised at $T_0 = 1$, and a resource stock $R(t)$ - over a fixed window of 1000 years discretized into yearly steps. During active steps technology grows linearly and resources deplete at a constant rate,

$$T_{t+1} = T_t + r, \quad R_{t+1} = R_t - \delta. \quad (1)$$

A collapse is triggered either by exhaustion ($R_t \leq 0$) or by a per-step Bernoulli existential hazard with rate h , evaluated independently of resource levels. A collapse multiplies technology by the survival fraction c_f , zeroes resources, and starts a dormant recovery period of rd years after which resources are restored to $rf \cdot R_0$. The *duty cycle* is the fraction of active years; the paper-style mean time to first collapse assigns the full window to runs that never collapse.

Table 1: Scenario anchor parameters (canonical transcription of Table 4, JANUS-PAPER-05).

Scenario	r	R_0	δ	c_f	rd (yr)	rf	h
S1	0.0020	400	1.2	0.6	40	0.15	0.003
S2	0.0026	700	1.5	0.3	70	0.1	0.001
S3	0.0001	2400	0	1	0	1	0
S4	0.0060	600	2.5	0.2	100	0.1	0
S5	0.0003	2500	0.1	0.9	20	0.8	0.0005
S6	0.0018	1600	1	0.4	50	0.25	0.005
S7	0.0060	400	1.5	0.3	60	0.2	0.001
S8	0.0015	800	1.3	0.6	70	0.35	0
S9	0.0011	900	1.3	0.5	70	0.35	0
S10	0.0002	2700	0	1	0	1	0

Monte Carlo replication follows the published uncertainty scheme (Table 5 of the same source): R_0 and δ normal with standard deviation 5% of the mean, c_f and rf triangular over mode $\pm 5\%$ clamped to $[0, 1]$, rd normal with a 10-year standard deviation truncated at zero, and r and h held fixed per scenario.

2 Replication of published outcomes

Each scenario is replicated twice: a 200-run batch seeded like the paper’s documented generator seed for direct comparability, and a 20000-run ensemble whose Monte Carlo error is small enough to expose systematic semantic differences. Table 2 compares ensemble aggregates against every prose-reported value; Table 3 lists each comparison with its tolerance. The overall verdict is pass. Shaded bands in Figure 1 show the tolerances; S3 and S10 must match exactly and do.

One residual divergence deserves note: the never-collapsed fraction of S5 is pinned by the model at $(1 - h)^{1000} = 0.6066$, while the paper reports roughly two thirds (≈ 0.66) from 200 runs. Binomial noise at that ensemble size covers most of the gap, so no semantic change was introduced to close it.

3 Censoring-aware survival analysis

The paper’s truncated mean conflates “collapsed late” with “never collapsed”. This experiment therefore treats never-collapsed runs as right-censored at year 1000 and reports Kaplan–Meier

Table 2: Ours / reported, by scenario (large ensembles). “—” marks values the source shows only in a figure.

Scenario	Duty cycle	First collapse (yr)	Collapse count	Never collapsed	Verdict
S1	0.631 / —	211 / —	9.72 / 10.00	0.000 / 0.000	pass
S2	0.611 / —	374 / —	5.95 / —	0.000 / 0.000	pass
S3	1.000 / 1.000	1000 / 1000	0.00 / 0.00	1.000 / 1.000	pass
S4	0.382 / 0.381	241 / —	6.63 / 6.60	0.000 / 0.000	pass
S5	0.990 / 0.990	786 / —	0.50 / —	0.607 / 0.660	pass
S6	0.793 / —	199 / —	4.28 / 4.50	0.006 / 0.000	pass
S7	0.580 / 0.570	234 / 223	7.38 / 7.50	0.000 / 0.000	pass
S8	0.873 / 0.870	618 / —	1.94 / 2.00	0.000 / 0.000	pass
S9	0.911 / 0.910	694 / 689	1.47 / 1.50	0.000 / 0.000	pass
S10	1.000 / 1.000	1000 / 1000	0.00 / 0.00	1.000 / 1.000	pass

survival, Nelson–Aalen cumulative hazard, and Aalen–Johansen cumulative incidence separating the two competing collapse causes (resource depletion vs. exogenous hazard). Figure 3 shows both; the cause decomposition recovers the expected physics - S6 is hazard-dominated, S4 collapses are almost purely resource-driven, and S1 mixes the two.

4 Global sensitivity analysis

Where the source sweeps one parameter at a time near scenario baselines, this experiment estimates variance decomposition over the whole seven-dimensional design space using Saltelli-style Sobol indices on independent uniform designs, replicated four times so every estimate carries a Monte Carlo standard error, plus Morris elementary effects in unit-cube coordinates (Figure 4).

Two structural findings act as internal consistency checks. First, the collapse fraction c_f and growth rate r have exactly zero total-order effect on every timing outcome - the model’s metrics depend only on event timing, which is precisely why the paper omits c_f from its sensitivity study. Second, the exact analytic recursions and the event-driven simulator evaluated under common random numbers produce index estimates that agree within their standard errors (columns 1 and 2 of Table 5).

By total-order effect the duty cycle is dominated by recovery delay rd and hazard h , followed by R_0 , then δ and rf ; expected collapse count is dominated by h (Table 5). Morris ranking agrees: rd , h , R_0 lead, with cf and r exactly zero. These rankings answer a different question than the source’s per-scenario swings - they describe variance across the entire design space, where interactions between parameters are active (first-order sums fall short of totals).

5 Level-2 continuous-parameter dataset

The ten scenarios anchor a continuous space; they are not training examples. Version jrl-dataset-1.0.0 contains 65,536 runs (8,192 design points $\times$ 8 seeds), sampled with a scrambled Sobol sequence over the paper-derived bounds in Table 6, executed by the event-driven sampler whose equivalence to the reference simulator is asserted by the test suite. Replicate structure keeps stochastic seed noise (mean within-point sd of 0.052 duty-cycle units) separable from parameter response (between-point spread 0.104).

Summary: mean duty cycle 0.868 (sd 0.120); mean collapse count 2.93; never-collapsed fraction 0.117; among collapsed runs, 85.0% hazard-triggered vs. 15.0% resource-depleted; regime counts

Table 3: Individual metric comparisons.

Scenario	Metric	Ours	Reported	Abs. diff.	Tolerance	Verdict
S1	fraction_never_collapsed	0.0000	0.0000	0.0000	0.080	pass
S1	mean_collapse_count	9.7162	10.0000	0.2838	1.200	pass
S2	fraction_never_collapsed	0.0000	0.0000	0.0000	0.080	pass
S3	fraction_never_collapsed	1.0000	1.0000	0.0000	0.000	pass
S3	mean_duty_cycle	1.0000	1.0000	0.0000	0.000	pass
S3	mean_time_to_first_collapse	1000.0000	1000.0000	0.0000	0.000	pass
S3	mean_collapse_count	0.0000	0.0000	0.0000	1.200	pass
S4	fraction_never_collapsed	0.0000	0.0000	0.0000	0.080	pass
S4	mean_duty_cycle	0.3817	0.3810	0.0007	0.050	pass
S4	mean_collapse_count	6.6346	6.6000	0.0346	1.200	pass
S5	fraction_never_collapsed	0.6073	0.6600	0.0527	0.080	pass
S5	mean_duty_cycle	0.9903	0.9900	0.0003	0.050	pass
S6	fraction_never_collapsed	0.0063	0.0000	0.0063	0.080	pass
S6	mean_collapse_count	4.2767	4.5000	0.2233	1.125	pass
S7	fraction_never_collapsed	0.0000	0.0000	0.0000	0.080	pass
S7	mean_duty_cycle	0.5802	0.5700	0.0102	0.050	pass
S7	mean_time_to_first_collapse	234.2971	223.0000	11.2971	45.000	pass
S7	mean_collapse_count	7.3768	7.5000	0.1232	1.200	pass
S8	fraction_never_collapsed	0.0000	0.0000	0.0000	0.080	pass
S8	mean_duty_cycle	0.8726	0.8700	0.0026	0.050	pass
S8	mean_collapse_count	1.9446	2.0000	0.0554	0.500	pass
S9	fraction_never_collapsed	0.0000	0.0000	0.0000	0.080	pass
S9	mean_duty_cycle	0.9112	0.9100	0.0012	0.050	pass
S9	mean_time_to_first_collapse	694.4534	689.0000	5.4534	45.000	pass
S9	mean_collapse_count	1.4670	1.5000	0.0330	0.375	pass
S10	fraction_never_collapsed	1.0000	1.0000	0.0000	0.000	pass
S10	mean_duty_cycle	1.0000	1.0000	0.0000	0.000	pass
S10	mean_time_to_first_collapse	1000.0000	1000.0000	0.0000	0.000	pass
S10	mean_collapse_count	0.0000	0.0000	0.0000	1.200	pass

7,669 stable / 10,375 single-collapse / 47,492 recurrent (Figure 6).

6 Surrogate benchmark under grouped holdouts

A learned model scoring well after a random split may only be rediscovering analytic structure. Every candidate is therefore compared against the exact renewal baseline under progressively harder protocols: a design-point-grouped random split; a contiguous sub-box ($\delta \geq 1.8$, $\text{rf} \leq 0.375$) held out entirely; and leave-one-anchor-out folds excluding all runs within normalized radius 0.45 of each in-domain scenario (seven folds; S3/S10 sit outside the design space). Split-conformal bands at nominal 90% are calibrated per fold, and empirical coverage is reported next to every score.

Tables 7 and 8 give the leaderboard as *point-level* R^2 (replicates averaged per design point, removing seed noise) with conformal coverage in parentheses. Figure 7 visualises the degradation pattern.

The honest headline: the analytic baseline wins everywhere, including the easy protocol, and learned models degrade under shift while coverage guarantees erode with it (the Gaussian process drops to 0.66 empirical coverage in the region-block fold). Any future claim that a learned surrogate beats this simulator must clear these protocols first.

7 Technosignature persistence and observation probabilities

The source defines, for technosignature i present in a scenario,

$$p_i = \{D_c + \min[T_i/T_{\text{span}}, 1 - D_c]\} \delta_i, \quad T_{\text{span}} = 1000 \text{ yr}, \quad (2)$$

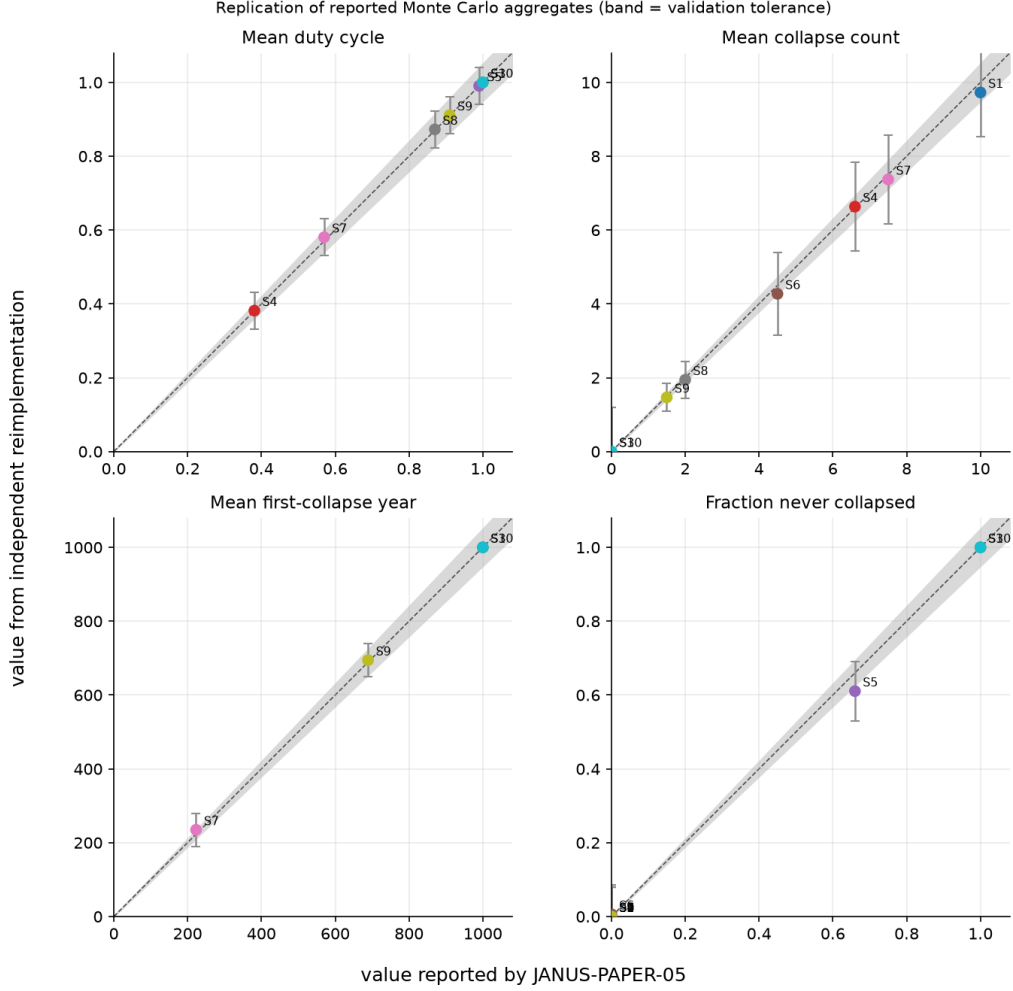


Figure 1: Reimplementation vs. reported aggregates. Error bars are validation tolerances; the dashed line is perfect agreement.

so short-lived signatures track the duty cycle while long-lived relic signatures remain observable through inactive periods. Lifetimes come from the paper’s own prose (NO₂ hours–days; CFC-11 55 yr; CFC-12 140 yr; CF₄ about 50 kyr). Presence flags are derived here from the reviewed canonical detection matrix (`janus.observability.apjl-figure-6`): NO₂ counts present where HWO detects CO₂+NO₂, CFCs where LIFE detects CO₂+CFC-11/12. CF₄ presence cannot be separated from that matrix and remains explicitly *not evaluated* instead of guessed. Table 9 and Figure 8 use this experiment’s reimplemented duty cycles; the committed report also contains the variant computed from canonical reported duty cycles where available.

8 Adjacent NASA context data

A contextual stellar table was retrieved live from the NASA Exoplanet Archive TAP service (24 confirmed planet-host systems within 15 pc with effective temperatures between 5300 and 6000 K, retrieved 2026-08-21). It is input for future observer-side work only and is not a canonical Janus dataset. Required acknowledgment, recorded verbatim in the snapshot:

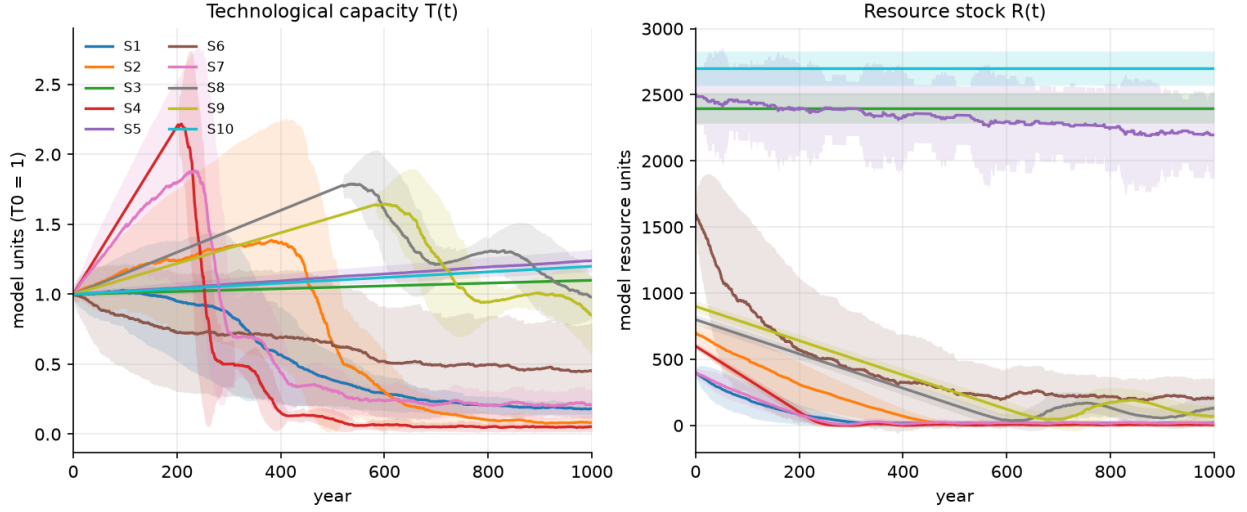


Figure 2: Ensemble trajectories of technological capacity and resources (200 runs per scenario, published perturbation scheme). The sawtooth pattern of S4, the late collapses of S8/S9, and the uninterrupted growth of S3/S10 reproduce the qualitative structure reported by the source.

This research has made use of the NASA Exoplanet Archive, which is operated by the California Institute of Technology, under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program.

9 Limitations and threats to validity

- Yearly-step semantics involve ambiguities the source does not pin down (same-step tie-breaking, whether the collapse year counts as active). Each choice is recorded in ADR-0001, implemented identically in all three representations, and covered by validation tolerances.
- Table 5 distributions are interpreted as normal standard deviations of 5% of the mean and symmetric triangular supports clamped to $[0, 1]$; the source does not specify clamping behaviour near distribution limits.
- S5’s never-collapsed fraction reproduces to 0.61 vs the reported ≈ 0.66 ; the analytic value is pinned at $(1 - h)^{1000} = 0.6066$ and the residual is consistent with 200-run Monte Carlo noise.
- First-order Sobol indices carry wide standard errors because the product estimator is heavy-tailed; total-order indices are the stable ranking and are always printed with standard errors.
- Leave-one-anchor-out folds exclude S3/S10 because $\delta = 0$ sits outside the continuous design space, whose lower bound follows the source’s own Table 6 sweep floor.
- The Gaussian process baseline is deliberately frugal (single optimizer restart, subsampled training); its scores bound naive GP usage rather than best-possible GP performance.

Table 4: Per-scenario survival summary (20k-run ensembles). RMST = restricted mean collapse-free time over the full window. The final two columns contrast the paper-style truncated first-collapse mean with the censoring fraction that motivates proper survival treatment.

Scenario	KM median (yr)	RMST (yr)	CIF _{res} (250)	CIF _{haz} (250)	Naive T_c	Censored
S1	233	212	0.000	0.526	212	0.000
S2	442	374	0.000	0.219	374	0.000
S3	—	1000	0.000	0.000	1000	1.000
S4	241	241	0.722	0.000	241	0.000
S5	428	455	0.000	0.306	782	0.600
S6	138	194	0.000	0.720	199	0.007
S7	260	235	0.143	0.218	235	0.000
S8	616	617	0.000	0.000	617	0.000
S9	693	694	0.000	0.000	694	0.000
S10	—	1000	0.000	0.000	1000	1.000

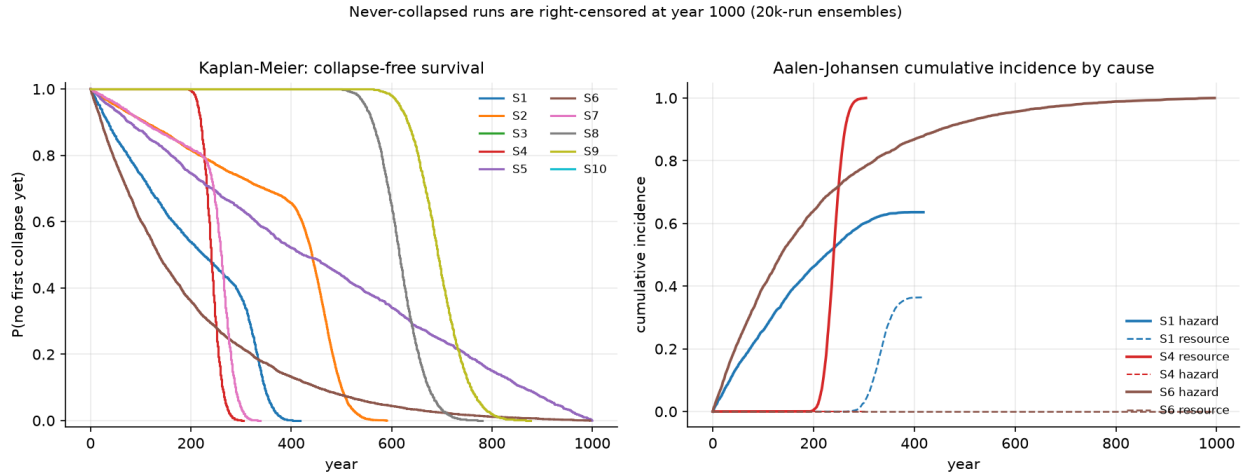


Figure 3: Left: Kaplan–Meier collapse-free survival per scenario. Right: Aalen–Johansen cumulative incidence for selected scenarios, split by cause.

10 Provenance and rights

- Model equations, parameters, uncertainty scheme, sweep bounds, and the persistence equation: JANUS-PAPER-05 (arXiv:2604.13774v1, CC BY 4.0), loaded at runtime from the reviewed canonical transcription data/canonical/collapse/model-and-reported-results.json.
- Presence flags derived from janus.observability.apjl-figure-6 (JANUS-PAPER-03, CC BY 4.0).
- The author-hosted **technocycles** repository is public but carries no license; it is cited, not copied. This reimplementaion was written from the paper’s equations alone.
- NASA Exoplanet Archive data carries the acknowledgment quoted in Section 8.
- No scenario probability, ranking, or forecast is produced anywhere in this pipeline.

Table 5: Total-order Sobol indices (mean $\pm$ standard error over four replicates; 1024 base samples).

Parameter	Analytic duty	Simulated duty (CRN)	Analytic count
r	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
R0	0.167 $\pm$ 0.011	0.250 $\pm$ 0.012	0.344 $\pm$ 0.035
delta	0.131 $\pm$ 0.003	0.218 $\pm$ 0.005	0.261 $\pm$ 0.015
cf	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
rd	0.501 $\pm$ 0.010	0.471 $\pm$ 0.010	0.112 $\pm$ 0.015
rf	0.107 $\pm$ 0.011	0.147 $\pm$ 0.010	0.262 $\pm$ 0.015
h	0.283 $\pm$ 0.019	0.350 $\pm$ 0.017	0.393 $\pm$ 0.017

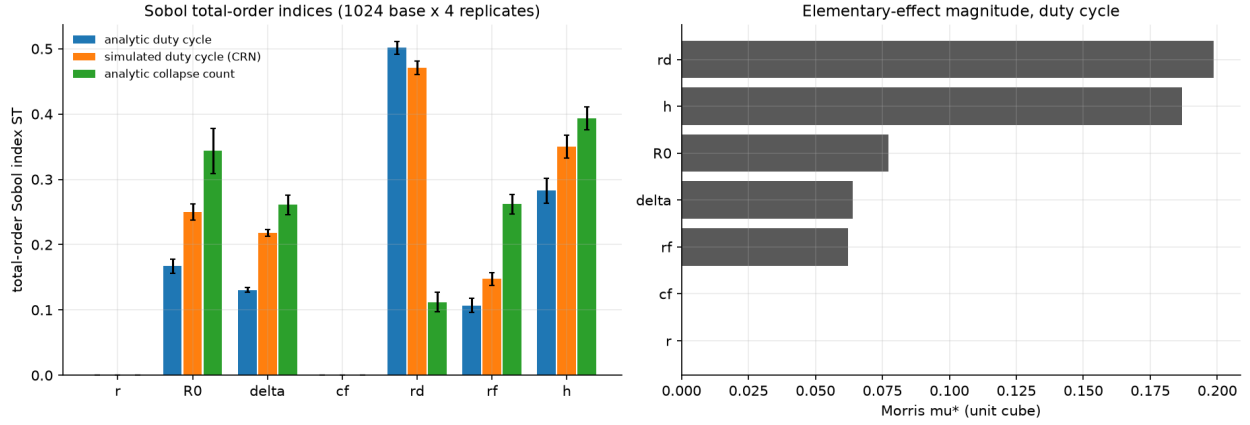


Figure 4: Left: total-order indices for three outputs. Right: Morris elementary-effect magnitudes on the unit cube.

Environment

Python 3.13.5, NumPy 2.5.2, SciPy 1.18.0, scikit-learn 1.9.0, captured 2026-08-22T00:03:24+00:00.
Git commit unavailable.

A Reproducing every number and figure

All artifacts regenerate deterministically from the repository state recorded in the report headers:

```
uv sync --project experiments/resilience-lab --extra dev --extra ml
uv run --project experiments/resilience-lab pytest experiments/resilience-lab/tests
uv run --project experiments/resilience-lab janus-resilience replicate
uv run --project experiments/resilience-lab janus-resilience survival
uv run --project experiments/resilience-lab janus-resilience sensitivity
uv run --project experiments/resilience-lab janus-resilience dataset
uv run --project experiments/resilience-lab janus-resilience surrogate
uv run --project experiments/resilience-lab janus-resilience persistence
uv run --project experiments/resilience-lab janus-resilience figures
uv run --project experiments/resilience-lab janus-resilience report
```

Every figure is re-rendered from committed JSON reports and seeded ensembles; every table in

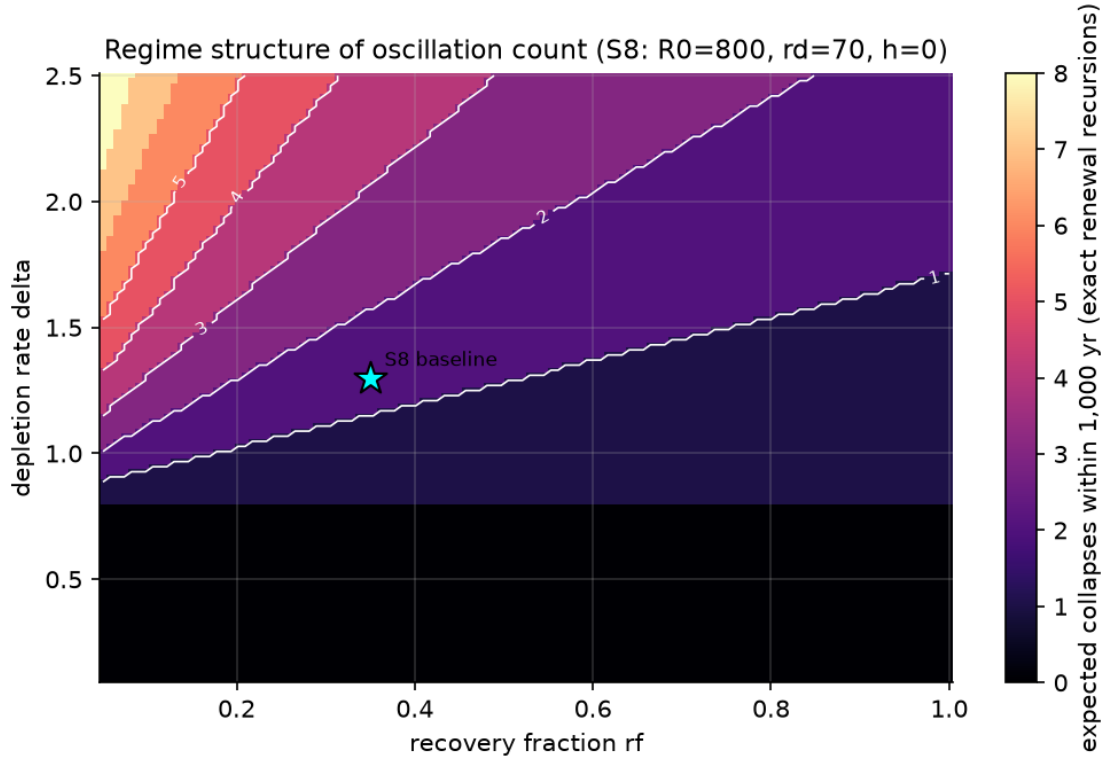


Figure 5: Expected number of collapse–recovery cycles across the rf – δ plane at S8 settings, computed from the exact renewal recursions rather than Monte Carlo. The S8 baseline sits near the boundary between the two- and three-cycle regimes, matching the source’s qualitative finding.

this document was generated directly from those JSON files, not transcribed.

Table 6: Design-space bounds and their provenance.

Parameter	Range	Source of bounds
r	[0.0001, 0.006]	anchor growth rates (scenario paper Table 9)
R0	[400, 2700]	anchor resource stocks
delta	[0.1, 2.5]	Table 6 sweep bounds
cf	[0.2, 1]	Table 6 joined with Table 2 governance ranges
rd	[0, 100]	Table 6 sweep bound; low end from Rule-by-All range
rf	[0.1, 1]	Table 6 low bound to full-recovery upper end
h	[0, 0.005]	Table 6 sweep bounds

Table 7: Duty-cycle target: point-level R^2 / empirical conformal coverage, by protocol. Negative R^2 means worse than predicting the test-set mean.

Model	random	region block	anchor out
analytic renewal	0.954 / 0.90	0.975 / 0.88	0.932 / 0.90
Gaussian process (RBF)	-0.001 / 0.87	-0.363 / 0.66	-7.862 / 0.81
hist gradient boosting	0.937 / 0.84	0.905 / 0.77	0.872 / 0.83
ridge (quadratic)	0.772 / 0.90	0.584 / 0.75	-2.295 / 0.85

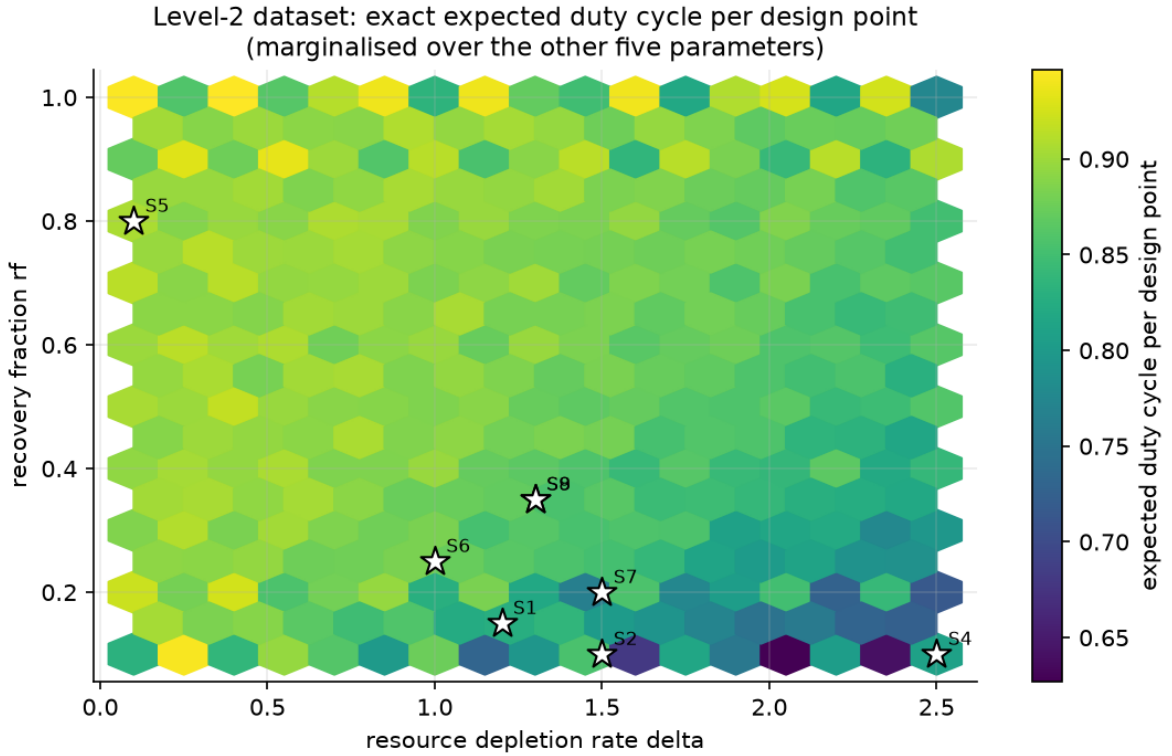


Figure 6: Exact expected duty cycle per design point, marginalised over the other five parameters. Scenario anchors sit inside the space except S3/S10, whose $\delta = 0$ lies on the excluded boundary.

Table 8: Collapse-count target: same layout.

Model	random	region block	anchor out
analytic renewal	0.942 / 0.89	0.974 / 0.85	0.858 / 0.91
hist gradient boosting	0.910 / 0.83	0.801 / 0.73	0.784 / 0.86
ridge (quadratic)	0.663 / 0.90	0.279 / 0.72	0.312 / 0.89

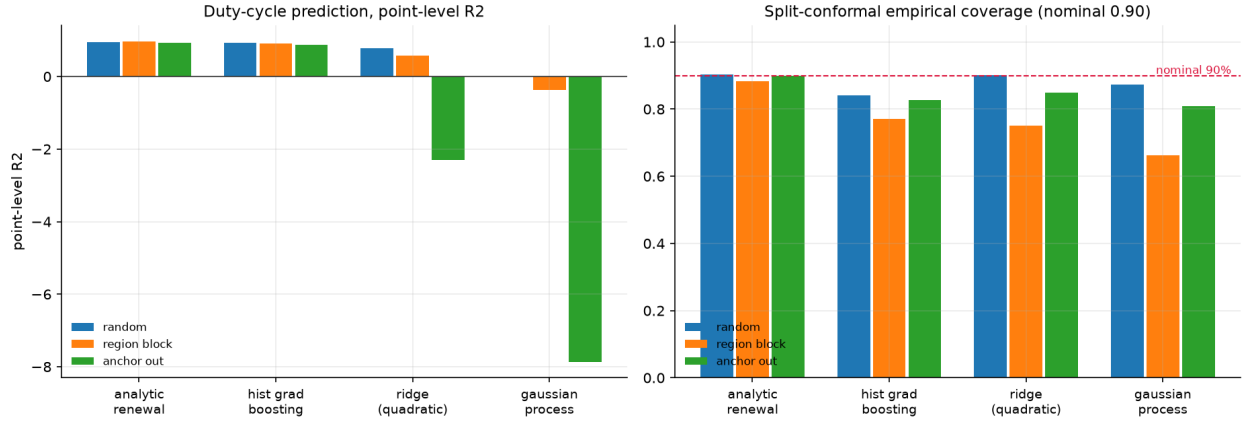


Figure 7: Benchmark degradation (left) and conformal coverage erosion (right). The exact analytic baseline loses almost nothing out-of-distribution because its residuals are irreducible seed noise rather than model error.

Table 9: Observation probabilities p_i from reimplemented duty cycles. A dash means the presence flag is zero in the canonical detection matrix.

Scenario	D_c used	NO2	CFC-11	CFC-12
S1	0.632	0.632	0.687	0.772
S10	1.000	0.000	0.000	0.000
S2	0.610	0.610	0.665	0.750
S3	1.000	1.000	1.000	1.000
S4	0.382	0.382	0.000	0.000
S5	0.990	0.000	0.000	0.000
S6	0.793	0.793	0.848	0.933
S7	0.580	0.580	0.000	0.000
S8	0.872	0.872	0.927	1.000
S9	0.911	0.000	0.000	0.000

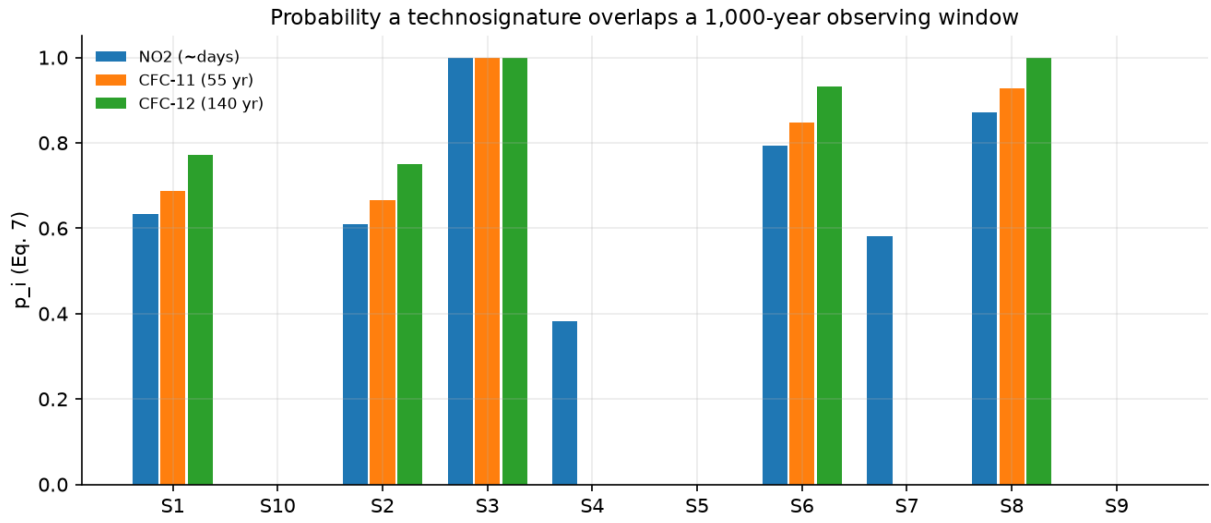


Figure 8: Short-lived signatures track the duty cycle exactly; longer-lived relics add observability during inactive periods. Scenarios lacking a canonical detection flag show nothing regardless of duty cycle - technology does not imply detectability.